Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Numerical Methods

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Explain the change of sign method for locating a root. **[2]**

2. Show that $f(x) = x^3 - x - 1$ has a root in $(1, 2)$. **[2]**

3. State the Newton-Raphson formula. **[2]**

4. Use Newton-Raphson with $x_0 = 1$ to find the next approximation to the root of $x^3 - x - 1 = 0$. **[4]**

5. State the formula for the trapezium rule with n strips. **[2]**

6. Use the trapezium rule with 4 strips to approximate $\int_0^1 e^{x^2}\,\mathrm{d}x$. **[4]**

7. State the conditions for a fixed-point iteration $x_{n+1} = g(x_n)$ to converge. **[3]**

8. Rearrange $x^3 - x - 1 = 0$ into the form $x = g(x)$ suitable for iteration. **[2]**

9. Perform two iterations of $x_{n+1} = \sqrt[3]{x_n + 1}$ starting at $x_0 = 1$. **[3]**

10. Explain what is meant by a staircase diagram for iteration. **[2]**

11. Use linear interpolation between $x = 1$ and $x = 2$ for $f(x) = x^3 - x - 1$. **[3]**

12. State Simpson's rule for n strips (n even). **[2]**

13. Approximate $\int_0^{\pi} \sin x \, \mathrm{d}x$ using Simpson's rule with 4 strips. **[4]**

14. Explain why the Newton-Raphson method may fail. **[2]**

15. Given $x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$, find the limit if the sequence converges. **[3]**

Section B – Hard (Q16–25)

16. Use Newton-Raphson to find a root of $x^3 - 3x + 1 = 0$ starting at $x_0 = 0$, to 3 d.p. **[5]**

17. Use the trapezium rule with 6 strips to approximate $\int_0^3 \frac{1}{1+x^2} \, \mathrm{d}x$ and compare with $\arctan 3$. **[5]**

18. Show that the iteration $x_{n+1} = e^{-x_n}$ converges for a suitable starting value and find the root of $x = e^{-x}$. **[5]**

19. Use Simpson's rule with 4 strips to approximate $\int_1^2 \frac{1}{x} \, \mathrm{d}x$ and compare with $\ln 2$. **[5]**

20. The equation $x = \cos x$ is solved by iteration. Investigate convergence of $x_{n+1} = \cos x_n$. **[4]**

21. Use Newton-Raphson to solve $e^{-x} = x$ starting at $x_0 = 0.5$, giving 3 iterations. **[5]**

22. Explain how to improve the accuracy of the trapezium rule. **[2]**

23. Find an approximate root of $x^3 + x - 1 = 0$ using binary search (bisection) with 4 steps from $[0, 1]$. **[5]**

24. Given $g'(x)$ at the fixed point, explain the cobweb diagram when $|g'| < 1$. **[3]**

25. Use the Newton-Raphson method to find $\sqrt{10}$ by solving $x^2 - 10 = 0$, starting at $x_0 = 3$. **[4]**

End of Worksheet • *Show all working clearly.*